

EXPLORATORY DATA ANALYSIS (EDA) SUMMARY REPORT

Project: GenAI-Powered Credit Delinquency Prediction Framework | **Client:** Geldium Finance | **Prepared by:** Meghana, AI Transformation Consultant (Tata iQ)

1. Introduction

The primary objective of this project is to support Geldium Finance in reducing its expanding credit card delinquency rates by building a robust, AI-powered predictive solution that replaces broad, historical rule-based segmentations. This Exploratory Data Analysis (EDA) Summary Report documents the critical baseline phase of data validation and analysis. The core goal here is to establish absolute data integrity, audit the quality of historical client performance information, and unearth structural patterns or risks before executing any predictive modeling steps.

2. Dataset Overview

A comprehensive validation structural scan was performed across the core credit accounts database provided by Geldium Finance to check variables, row alignments, and formats.

- **Number of records:** 500 complete customer rows (ranging indices 0 to 499).
- **Key variables:** 19 columns mapping unique attributes including operational identifiers (`Customer_ID`), customer demographics (`Age` , `Employment_Status` , `Location`), explicit risk parameters (`Income` , `Credit_Score` , `Credit_Utilization` , `Missed_Payments` , `Loan_Balance` , `Debt_to_Income_Ratio` , `Account_Tenure` , `Credit_Card_Type`), and 6 chronological payment tracking windows (`Month_1` down through `Month_6`).
- **Data types:** Distributed cleanly across 5 continuous floating-point fields (float64), 4 structured integers (int64), and 10 categorical string variables (object).

3. Missing Data Analysis

Identifying and addressing missing data fields is an imperative procedural task to maintain structural symmetry and insulate machine learning algorithms against structural prediction skew. Below details the missing features identified along with their prescriptive treatment regimes:

Variable Name	Missing Count	Missing (%)	Prescribed Mitigation & Justification
Income	39 records	7.8%	Median Imputation segmented by <code>Employment_Status</code> . Protects the data volume while preventing extreme income outliers from altering statistical weights.
Loan_Balance	29 records	5.8%	Overall Median Imputation. Prevents severe loan portfolio debt outliers from misrepresenting typical customer exposure metrics.
Credit_Score	2 records	0.4%	Mean Imputation. Because the volume of missing elements is negligible, a basic mean replacement preserves normal distribution.

4. Key Findings and Risk Indicators

A direct analysis matching customer behavioral features to historical delinquency flags exposed critical risk signals that will form the logical backbone of the future model architecture.

- **Correlations observed between key variables:** Data distributions indicate a severe correlation spike when financial constraints compress. Specifically, customers carrying a **Debt-to-Income (DTI) ratio exceeding 0.50** combined with high **Credit Utilization levels** display immediate correlation alignment to the target `Delinquent_Account` flag, highlighting critical leverage stress.
- **Unexpected anomalies:**
 1. **Zero Tenure Discrepancy:** 28 customer rows show `Account_Tenure == 0` but contain extensive historical flags under `Month_1` , marking clear database configuration errors.
 2. **Income Contradiction:** 84 profiles flagged explicitly as "Unemployed" contain high active monthly income logs crossing \$10,000.
 3. **Utilization Bound Breach:** 4 accounts display a physical impossibility with `Credit_Utilization > 1.0` (exceeding maximum credit boundaries).

5. AI & GenAI Usage

Generative AI technologies were integrated dynamically within this exploratory sequence to augment rapid syntax creation, organize programmatic cleaning templates, and outline industrial standard imputation patterns securely.

Documented GenAI Prompts Utilized:

- *"Write a non-destructive Pandas validation loop script to map logical matrix contradictions between zero customer tenure fields and filled historical payment columns."*
- *"Design a robust mathematical justification framework for treating skewed continuous income distributions with missing rows without reducing dataset length."*

6. Conclusion & Next Steps

The exploratory verification confirms that the data contains highly predictive behavioral patterns needed to construct a strong risk-mitigation tool. However, internal systemic anomalies and empty spaces must be corrected first to achieve unbiased modeling accuracy.

Prescribed Operational Roadmap:

- Execute conditional median and mean imputations according to the Step 3 framework.
- Correct mathematical anomalies by capping maximum credit utilization fields to a hard ceiling of 1.0 (100%) and separating the 84 contradictory unemployed high-income records for custom validation.
- Transition clean structures forward into feature-engineering and predictive machine learning stages.